GEN-26—Modifications to approved weighing and measuring devices and electricity and gas meters

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1.0 Purpose

The purpose of this bulletin is to provide Measurement Canada's (MC) requirements for reporting modifications to the design, composition or construction of approved devices.

2.0 Scope

This bulletin applies to weighing and measuring devices, and to electricity and gas meters which have been approved for use in trade by MC pursuant to the *Weights and Measures Act* or the *Electricity and Gas Inspection Act*. For the purpose of this bulletin, weighing and measuring devices, and electricity and gas meters will be referred to as devices.

3.0 Background

A Notice of Approval (NOA) issued under sections 14 and 15 of the *Weights and Measures Regulations* or sections 13 and 14 of the *Electricity and Gas Inspection Regulations* applies to any device that is, in every material respect, identical to the device originally submitted for approval evaluation. During the lifespan of a device, the manufacturer may decide to make design modifications that may be superficial, administrative in nature, or can affect the metrological or technical characteristics of the device. Some changes may affect the approval status of the new device design.

4.0 Reporting modifications to approved devices

4.1 Subject to clauses 4.2 to 4.5, modifications to a device that alter the originally approved design shall be reported by the manufacturer or approval applicant to MC's Engineering and Laboratory Services Directorate (ELSD). The ELSD will review the proposed changes and advise the manufacturer or approval applicant if the modifications warrant a re-evaluation of the device and/or modification of the original NOA. The ELSD may request additional information to assess the nature of the modification. If the modification or modifications are clearly identifiable and are not deemed to have any apparent effect on the other metrological parameters, then only a partial evaluation of the affected characteristics may be necessary to determine compliance with applicable MC requirements currently in force.

> **Note:** When modifying an approved device, the manufacturer should note that the modifications may not be covered by the current NOA. In these cases, an amendment to the existing approval or a new approval may be required.

4.2 No formal notification to the ELSD is required when a modification to an approved design clearly represents an insignificant or superficial change that will not affect administrative, technical or metrological elements of the device (e.g., colour of parts). Whenever the significance of such a modification is uncertain, applicants should contact the ELSD for assistance in this determination.

4.3 Modifications that affect more significant features related to the design, composition and construction of a device but do not have an impact on metrological characteristics (e.g., case style, layout, etc.) shall be reported to the ELSD in order to maintain an accurate description of the approved device. These modifications are considered to be metrologically

insignificant and the modified device is still covered by the original NOA. In such cases, the ELSD may choose to issue a Modification Acceptance Letter (MAL) to convey information about modifications of a device. These types of modifications can often be assessed by a review of the documentation referred to in clause 4.5.

> **Note:** For electricity and gas devices that are now subject to MC's software specifications in S-EG-05, MALs shall no longer be issued by the Agency to communicate device software update capability.

4.4 In addition to clause 4.3, where external configuration and interrogation software (used to perform functions such as the configuration of legally relevant parameters, reading device information, applying electronic security provisions, etc.) has been specifically identified in the NOA, any revisions to such software shall be reported to the ELSD. Where proprietary communication or configuration devices have been specifically identified in the NOA, any modification to such devices shall also be reported to the ELSD.

4.5 A Notice of CLASS Approval (NOCA) will identify the CLASS approval holder. The authority to manufacture and install devices using a NOCA extends only to the CLASS approval holder. Use of the CLASS approval, including significant modifications to existing CLASS approved weighing devices, by persons other than the CLASS approval holder shall be approved in writing by the CLASS approval holder.

4.6 An approval modification request submitted to MC shall include a description of the changes that have been made to the device and shall originate from the original approval applicant.

> **Note:** Information submitted by an applicant during the device type approval process is considered as proprietary information that will only be reviewed with the original applicant.

5.0 Assessing modifications to approved devices—general examples

With the ever-increasing variety and complexity of devices, the distinction between what constitutes a metrologically significant and insignificant modification is not readily apparent. As it would be impractical to describe and list all the possible types of modifications that can be made to a device, only the most common types of modifications have been presented in this section for the purpose of assessment.

5.1 Metrologically significant hardware modifications

Metrologically significant hardware modifications include changes to the composition, construction and design that may affect the performance of a device. Modifications of this nature can increase the potential for fraudulent use of a device or compromise a device's ability to measure accurately within its expected life cycle, seal period, or periods of time otherwise prescribed by regulations and specifications. Examples of such modifications include the following:

a. Modifications to the material composition of any part or component of an approved device causing it to be substantially different than the original composition, which can result in premature wear, increased measurement inaccuracy, or incompatibilities with other components of the device or with the commodity that it is intended to measure.

b. Modifications to the construction of an approved device which compromise its ability to resist damage through exposure to the operating conditions, as may normally be encountered during its intended service.
c. Modifications to the original approved design of a device which include the reconfiguration, replacement, and removal or addition of components and ancillary equipment that were not originally included in the approved design and can affect the measurement accuracy of the device.

5.2 Legally relevant software modifications that affect metrological functions

5.2.1 Legally relevant software is the software, or part of the software, that interferes with properties regulated by legal metrology. Some examples of legally relevant software modifications include:

i. configuration parameters that are either adjustable or selectable by the user and can affect the accuracy of measurement or resulting transaction, or increase the potential for fraudulent use of a device;
ii. computational functions where gross measured values or stored information are used in computations that result in net measured values or extended prices;
iii. control functions where measured values and/or other system inputs and outputs are used to control measurement processes and data acquisition related to the measurement process;
iv. measurement functions that could comprise any part of a measuring sequence, from the sensing of the commodity or service to be measured, up to and including the display of the results of the measurement;

v. changes to operating system software, device drivers, interfacing program modules, and boot programs;
vi. modifications to metrological audit trail functions and their method of access.

5.2.2 In accordance with clause 4.1, any software modifications that affect metrological functions of the device shall be reported to the ELSD. A metrological function is defined as a function of a device that is necessary for the measurement process, including the sensing of the measured quantity as well as the transmission, processing, storage and correction or adjustment of measurement signals or values that can affect:

a. the final quantity determination or price calculation, as applicable; or
b. the validity of the transactions.

6.0 Revisions

The purpose of this revision is to make additional clarifications to clause 4.4 regarding the reporting of modifications to approved devices, in particular devices using external configuration or interrogation software.

The purpose of revision 2 (2015-05-13) was to make the bulletin more concise and clarify the reporting of modifications to approved devices, in particular devices using external configuration or interrogation software.

The purpose of revision 1 (2008-09-01) was to simplify the requirements regarding modifications to approved weighing and measuring devices and electricity and gas meters.

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